

Generation Interconnection Request System Impact Study Report

For Queue Project Q12-0142

James E. Rogers Energy Complex 138 kV
August 29, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0142. The Interconnection Customer proposes a 751.0 MW natural gas combined cycle facility interconnecting at the James E. Rogers Energy Complex 138 kV substation in the Pennsylvania area. The request was received on July 10, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 92 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$142,966,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
COLLIERVILLE - JAMES E. ROGERS ENERGY COMPLEX 138.0 kV	NORWAY PT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138.0 kV	376.5	225	100.0
NORWAY PT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138.0 kV	COLLIERVILLE - JAMES E. ROGERS ENERGY COMPLEX 138.0 kV	371.4	228	100.0
BEARDSLEY - NORWAY POINT HYDROPOWER PROJECT 138.0 kV	JAMES E. ROGERS ENERGY COMPLEX - COLLIERVILLE SUBSTATION 138.0 kV	357.8	235	100.0
HIGH ST - LYNBROOK 138.0 kV	NORWAY PT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138.0 kV	222.7	201	45.4
FAIR OAKS - LYNBROOK 138.0 kV	NORWAY POINT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138.0 kV	222.4	200	45.4
HIGH ST - COON RAPIDS II 138.0 kV	NORWAY PT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138.0 kV	220.3	205	45.4
BAKEOVEN - BEARDSLEY 138.0 kV	COLLIERVILLE - JAMES E. ROGERS ENERGY COMPLEX SUBSTATION 138.0 kV	214.1	261	72.0
NOLENSVILLE - COON RAPIDS II 138.0 kV	JAMES E. ROGERS ENERGY COMPLEX - NORWAY POINT HYDROPOWER PROJECT 138.0 kV	189.1	240	45.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
NOLENSVILLE - SWEET 138.0 kV	JAMES E. ROGERS ENERGY COMPLEX - NORWAY PT HYDROPOWER PROJECT 138.0 kV	189.0	242	45.4
COLLIERVILLE - SWEET 138.0 kV	NORWAY POINT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138.0 kV	186.7	247	45.4
COLLIERVILLE - MERRILL HILLS 138.0 kV	NORWAY POINT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138.0 kV	168.6	228	54.6
7TH AVENUE N - CIMGEN 138.0 kV	BEARDSLEY - BAKEOVEN 138.0 kV	168.2	200	40.2
WOLFSKILL TAP SUBSTATION - MERRILL HILLS 138.0 kV line	NORWAY PT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138.0 kV	165.2	231	54.6
MANCHACA - CIMGEN 138.0 kV line	BEARDSLEY - BAKEOVEN 138.0 kV Line	161.7	210	40.2
BARNUM - WOLFSKILL TAP 138.0 kV line	NORWAY PT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138.0 kV Line	161.7	235	54.6
FAIR OAKS 345-138 kV TX	NORWAY POINT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138.0 kV	160.3	267	45.4
BARNUM - ALCONA 138.0 kV L	JAMES E. ROGERS ENERGY COMPLEX - NORWAY PT HYDROPOWER PROJECT 138.0 kV Line	156.6	239	54.6
BAKEOVEN SUBSTATION - N ALLEGHENY WINDPOWER PROJECT 138.0 kV L	COLLIERVILLE - JAMES E. ROGERS ENERGY COMPLEX 138.0 kV Line	154.9	255	51.5
HIGHLAND WIND PROJECT PA - ALCONA 138.0 kV	NORWAY POINT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138.0 kV	152.1	242	54.6
HIGHLAND WIND PROJECT PA - KENNECOTT 138.0 kV	NORWAY PT HYDROPOWER PROJECT - JAMES E. ROGERS EERGY COMPLEX 138.0 kV	148.9	246	54.6
MANCHACA - O'BANION 138.0 kV	BEARDSLEY - BAKEOVEN 138.0 kV L	138.6	218	40.2
O'BANION - PURDUE NORTHWEST 138.0 kV line	BEARDSLEY - BAKEOVEN 138.0 kV	135.6	227	40.2
PURDUE NORTHWEST - BEARDSLEY 138.0 kV	BEARDSLEY - BAKEOVEN 138.0 kV	134.7	231	40.2
BON AQUA - DRY GAP 138.0 kV	JAMES E. ROGERS ENERGY COMPLEX - NORWAY POINT HYDROPOWER PROJECT 138.0 kV	122.4	383	6.2
BON AQUA - CARL BAILEY 138.0 kV	NORWAY PT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138.0 kV Line	121.8	390	6.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
COLLIERVILLE - JAMES E. ROGERS ENERGY COMPLEX 138.0 kV	Rebuild line with higher-capacity conductor	\$5,822,000	88% (\$5,145,000)
NORWAY PT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138.0 kV	Rebuild line with higher-capacity conductor	\$8,188,000	88% (\$7,237,000)
BEARDSLEY - NORWAY POINT HYDROPOWER PROJECT 138.0 kV	Rebuild line with higher-capacity conductor	\$4,093,000	88% (\$3,618,000)
LYNBROOK - HIGH ST SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$2,726,000	88% (\$2,409,000)
FAIR OAKS - LYNBROOK 138.0 kV	Rebuild line with higher-capacity conductor	\$6,683,000	88% (\$5,907,000)
COON RAPIDS II - HIGH STREET 138.0 kV	Rebuild line with higher-capacity conductor	\$3,655,000	88% (\$3,231,000)
BAKEOVEN - BEARDSLEY 138.0 kV	Rebuild line with higher-capacity conductor	\$7,515,000	86% (\$6,464,000)
NOLESVILLE - COON RAPIDS II 138.0 kV	Rebuild line with higher-capacity conductor	\$22,460,000	88% (\$19,852,000)
NOLENSVILLE - SWEET 138.0 kV	Rebuild line with higher-capacity conductor	\$8,083,000	88% (\$7,144,000)
COLLIERVILLE - SWEET 138.0 kV	Rebuild line with higher-capacity conductor	\$2,901,000	88% (\$2,564,000)
COLLIERVILLE - MERRILL HILLS 138.0 kV	Rebuild line with higher-capacity conductor	\$6,019,000	88% (\$5,320,000)
7TH AVENUE N - CIMGEN 138.0 kV	Rebuild line with higher-capacity conductor	\$3,915,000	81% (\$3,189,000)
WOLFSKILL TAP SUBSTATION - MERRILL HILLS 138.0 kV line	Rebuild line with higher-capacity conductor	\$6,578,000	88% (\$5,813,000)
CIMGEN - MANCHACA 138.0 kV	Rebuild line with higher-capacity conductor	\$1,839,000	81% (\$1,498,000)
BARNUM - WOLFSKILL TAP 138.0 kV line	Rebuild line with higher-capacity conductor	\$7,317,000	88% (\$6,467,000)
FAIR OAKS 345/138kV Auto	Replace transformer with higher-rated unit	\$23,592,000	88% (\$20,853,000)
BARNUM - ALCONA 138.0 kV L	Rebuild line with higher-capacity conductor	\$8,645,000	88% (\$7,640,000)
BAKEOVEN - N ALLEGHENY WINDPOWER PROJECT 138.0 kV	Rebuild line with higher-capacity conductor	\$1,525,000	86% (\$1,312,000)
HIGHLAND WIND PROJECT PA - ALCONA 138.0 kV	Rebuild line with higher-capacity conductor	\$2,035,000	88% (\$1,799,000)
HIGHLAND WIND PROJECT PA - KENNECOTT 138.0 kV	Rebuild line with higher-capacity conductor	\$2,118,000	88% (\$1,872,000)
MANCHACA - O'BANION 138.0 kV	Rebuild line with higher-capacity conductor	\$7,602,000	85% (\$6,493,000)
O'BANION - PURDUE NORTHWEST 138.0 kV line	Rebuild line with higher-capacity conductor	\$6,859,000	85% (\$5,859,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BEARDSLY - PURDUE NORTHWEST 138.0 kV	Rebuild line with higher-capacity conductor	\$5,959,000	85% (\$5,090,000)
DRY GAP SUBSTATION - BON AQUA 138.0 kV	Rebuild line with higher-capacity conductor	\$5,566,000	60% (\$3,366,000)
BON AQUA - CRL BAILEY 138.0 kV	Rebuild line with higher-capacity conductor	\$4,670,000	60% (\$2,824,000)
	Total allocated to project		\$142,966,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.